

Excel Fundamentals Documentation

Module: Excel Basics for Data Processing

Learning Objectives

By the end of this session, learners will be able to:

- Navigate the Excel interface confidently
 - Apply number formatting
 - Use basic formulas and functions
 - Understand relative and absolute references
-

A. Navigating the Excel Interface

1 What is Excel?

Microsoft Excel is a spreadsheet application used for:

- Data entry
 - Data processing
 - Calculations
 - Analysis
-

2 Main Components of Excel Interface

◆ Ribbon

The Ribbon contains tabs and commands.

Important Tabs:

Tab	Purpose
Home	Formatting and editing
Insert	Charts, tables, images
Page Layout	Page settings
Formulas	Functions and formulas
Data	Sorting, filtering, validation
Review	Spell check and comments
View	Workbook views

◆ Workbook

A workbook is an Excel file.

Example:

None

Hospital_Data.xlsx

◆ Worksheet

A worksheet is a single sheet inside a workbook.

Default names:

- Sheet1
 - Sheet2
-

◆ Rows and Columns

Component	Example
Row	1,2,3
Column	A,B,C
Cell	A1, B2

◆ Cell

Intersection of row and column.

Example:

None
B3

◆ Range

Group of cells.

Example:

None
A1:D10



Practice Activity

1. Open Excel
2. Create a workbook
3. Rename Sheet1 to:

None
Hospital_Data

4. Enter sample data

B. Basic Cell and Range Formatting

1 Font Formatting

Located in:

None

Home → Font

You can:

- Change font style
 - Increase/decrease size
 - Bold
 - Italic
 - Underline
-

Example

Before

After

revenu REVENUE
e

Shortcut:

None

Ctrl + B → Bold

2 Cell Colors

Fill Color

Used to highlight cells.

Example:

- Revenue → Green
 - Pending → Red
-

3 Borders

Borders improve readability.

Options:

- All Borders
 - Thick Border
 - Bottom Border
-

4 Alignment

Used for arranging text.

Alignment	Purpose
Left	Default text
Center	Headings
Right	Numbers

5 Merge & Center

Combines cells.

Example:

None

A1:E1

Used for titles.



Practice Activity

Create this heading:

A1:E1

Hospital Revenue
Report

Apply:

- Bold
 - Center
 - Blue fill color
 - White font
-

C. Number Formatting

1 Currency Format

Used for:

- Salary
- Revenue
- Bills

Example:

None

₹ 5,000

Steps:

None

Home → Number → Currency

2 Percentage Format

Example:

None

85%

Used in:

- Growth rates
 - Attendance
 - Profit %
-

3 Date Format

Example:

None

07-May-2026

Used for:

- Invoice date
- Joining date

- Visit date
-

4 Custom Formatting

Example:

None
0001

Converts:

None
1 → 0001

Common Custom Format Codes

Format Code	Result
₹#, ##0	₹25,000
0%	85%
dd-mm-yyyy	11-05-2026
dd mmmm yyyy	11 May 2026

0.00

25.50

#, ##0

10,000

Useful for:

- Employee IDs
- Invoice numbers



Practice Activity

Format:

Data Type	Format
Revenue	Currency
Growth	Percentage
Joining Date	Date

D. Introduction to Basic Formulas and Functions

1 What is a Formula?

Formula begins with:

None

=

Example:

None

=A1+B1

2 SUM Function

Adds values.

Syntax:

None

=SUM(A1:A10)

Example:

None

Total Revenue

3 AVERAGE Function

Finds average value.

Syntax:

None

=AVERAGE(B1:B10)

Example:

None

Average Salary

4 COUNT Function

Counts numeric cells.

Syntax:

None

=COUNT(C1:C20)

5 MIN Function

Returns smallest value.

Syntax:

None

=MIN(D1:D20)

6 MAX Function

Returns largest value.

Syntax:

None

=MAX(D1:D20)



Practice Dataset

Employee	Salary
----------	--------

Ravi	25000
------	-------

Anita	30000
-------	-------

John	28000
------	-------

Try:

- Total Salary
- Average Salary
- Highest Salary
- Lowest Salary

E. Relative and Absolute References

1 Relative Reference

Changes automatically when copied.

Example:

None

=A1*B1

If copied downward:

None

=A2*B2

2 Absolute Reference

Does NOT change when copied.

Uses:

None

\$

Example:

None

=A1*\$D\$1

◆ Why Use Absolute Reference?

Example:

Tax Rate stored in:

None

D1 = 18%

Formula:

None

=B2*\$D\$1

When copied:

- D1 remains fixed
-



Practice Activity

Create:

Product Price Tax Rate Final Price

Use:

None

`=Price * $TaxRate$`

End of Session Assignment

Task:

Create a small employee dataset and apply:

- Formatting
- Currency format
- Date format
- SUM
- AVERAGE
- MIN/MAX
- Relative & absolute references